

AAHIL ALI

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EDUCATION

Cornell University

Expected May 2027

Major: Bachelor of Science in Mechanical Engineering, Dyson Business Minor for Engineers

Relevant Courses (*Currently Enrolled): Mechatronics*, Heat Transfer*, Propulsion of Aircrafts & Rockets*, Automotive Engineering*, Mechanics of Engineering Materials, System Dynamics, Fluid Dynamics, Thermodynamics, Dynamics

WORK EXPERIENCE

General Dynamics Electric Boat | Design & Engineering Co-op

Aug. 2026 – Present | Groton, CT

- Support engineering and procurement execution for COLUMBIA-class SSBNs, working directly with vendors to resolve fabrication and manufacturability challenges alongside conformance to procurement specifications and PO requirements.
- Review and disposition Vendor Information Requests by validating supplier technical data against engineering drawings, governing welding specifications, and 3D design models in Teamcenter.

Cornell Electric Vehicles Project Team | Steering Subteam

Oct. 2023 - Present | Ithaca, NY

- Collaborate with a group of 75+ to design and build a hyper-efficient autonomous EV for the Shell Eco-marathon, coming in 5th on the track and winning two off-track awards and \$4500 from the Shell Eco-marathon Americas competition.
- Designed and manufactured a lightweight acceleration pedal and sensor mounts in Autodesk Inventor; analyzed static forces in MATLAB to achieve 25 lbs of pedal feedback force; integrated a linear potentiometer to record position data for testing.
- Led steering system layout for 2026 vehicle and designed a threaded aluminum mounting plate that defines the interface for the rack and pinion, pedal box, and autonomy hardware, engineered for maximum stability and minimal weight; machined 50+ custom components using CNC, manual mill, and lathe to support vehicle assembly.

Lefiell Manufacturing | Manufacturing Engineering Intern

Summer 2026 | Santa Fe Springs, CA

- Developed Autodesk Inventor 3D models and process drawings for five Boeing main landing gear tubular components, translating customer drawings into machining operation sequences, intermediate part geometry, datum schemes, defining fixturing and work holding requirements, and applying GD&T per ASME Y14.5; all five parts released to production.
- Designed 10 custom thread plugs for RH/LH UNJF-3B threads per AS8879; created Autodesk Inventor assemblies for fit checks to verify thread engagement and drilling clearance, and iterated tooling geometry with CNC programmers.
- Created a gundrill depth gauge for dimensional verification of internal radius tangent point location on three tubular parts, using axially spaced gauging ridges matched to varying bore depths and consolidating three inspection gauges into one.
- Reverse engineered a main landing gear door forward stop fitting from a physical component and legacy Boeing hand drawing, using caliper measurements to reconstruct part geometry and model it in Autodesk Inventor for manufacturing.
- Validated sonobuoy launch components by cross checking CATIA V5 customer models against Autodesk Inventor models and engineering drawings, verifying dimensional accuracy, hole locations, and feature geometry before production.

Information & Decision Science Lab | Undergraduate Researcher

Aug 2025 – May 2026 | Ithaca, NY

- Contributing to the MathWorks-sponsored IDS³C Scaled Smart-City Testbed Project, a 1:25-scale research platform to test and validate autonomous vehicle control strategies in urban traffic systems.
- Implemented Arduino/Raspberry Pi hardware across 45 RC vehicles to enable scalable fleet testing and shorten debug cycles by 40%; created an engineering manual covering hardware/software integration & chassis updates, cutting onboarding time.

Cornell University MAE 2250 Mechanical Design | Teaching Assistant

Jan 2026 – May 2026 | Ithaca, NY

- Taught a weekly laboratory class of 25–35 undergraduate students on the topics of Computer-Aided Design, Machining, 3D Printing, Laser Cutting, Design for Manufacturing, Rapid Prototyping, plastic and metal part design, and other mechanical synthesis skills.

HydroClip

Jan 2025 – May 2025 | Ithaca, NY

- Created a backpack strap mounted water bottle holder; designed a sliding lock and latch geometry in Fusion 360, set clearances and tolerances for a secure fit, and iterated design with six 3D printed prototypes
- Conducted bench testing with pull, shake, and walk tests to adjust snap fit assembly and durability; refined latch geometry and strap fit to maximize strength in the direction of bending.

SKILLS

Technical: CAD (Inventor, Fusion, AutoCAD, SolidWorks), DFMA, GD&T, DFMEA, Master Modelling, Tolerance Stackup Analysis, ANSYS FEA (Mechanical, Fluent CFD), MATLAB, Arduino, Python, Excel, Office, PLEX MES, Teamcenter PLM

Manufacturing: Machining (Mill, Lathe, CNC), 3D Printing, Carbon Fiber (Vacuum Infusion, Wet Layup, Forged), Hand Tools

Information & Decision Science Lab | Undergraduate Researcher

Aug 2024 – Present | Ithaca, NY

- Contributing to the MathWorks-sponsored IDS³C Scaled Smart-City Testbed Project, a 1:25-scale research platform to test and validate autonomous vehicle control strategies in urban traffic systems
- Implemented Arduino/Raspberry Pi hardware across 45 RC vehicles to enable scalable fleet testing and shorten debug cycles by 40%; created an engineering manual covering hardware/software integration & chassis updates, cutting onboarding time by 50%.
- Built and tested autonomous driving features by integrating scaled road network models in MATLAB, wiring and calibrating force-feedback steering hardware, and driver controls for real-time testing

Created a gundrill depth gauge for dimensional verification of internal radius tangent point location on three tubular components, using axially spaced gauging ridges matched to varying bore depths and reducing inspection tooling from three gauges to one.

okay i want to update my resume for my summer internship and so what would be the best way to do that

it was a lefiell manufacturing an aerospace component manufacturing company and they mainly focus on tubular aerospace components

i deisgned 2 tubular aerospace parts supporting boeing which is one of their customers which was used for landing gear

they do all of their manufactruing in house

so i worked with machine operate to create process drawings to shwo each step of the maunfactrung process from facing boring and tapping ID to contour turing and flipping boring and tapping 2nd end and millings flats

i was worked with director of manufactruing to develop process routing for these parts and based on the processing routing i made rpocess darngs in Inventor

as you can imagine wince this was going to machine operators and the customer is boeing for landing gear, i working with toleralcing, gd&t principles, and ensuring that my drawings proper so that a machine operator could look and manufcating based on them

emphasize gd&t especially because it was a big part of when i was makig the process drawigs

the next thing i worked on was making tooling for these tubular component which were external thread plug gauages which could be used The purpose of a thread plug gauge for tubular components is to inspect and verify the accuracy of internal threads (such as those inside pipes, couplings, or threaded tube ends) to ensure proper fit, leak resistance, and structural integrity (i got that from online)

so for these tubular compliments i made I made 5 different thread plug gayages to fit a all of the different parts that we needed them for i amattaching photos

and these were made i according for to be manufactured on Mori Seiki Mill Turn Machine

also i was exposed to a variety of different manufacturing processes from swagig, to gundrill, ro flow, id machining, heat treating methods, its all in pictures i uploaded use them if they are helpful

I also made a another custom tooling which was for a gundrill depth gauage so we have parts that need gundril and we need to check the ID of the gundrill parts to ensure it is correct and so i made a singular tool that could be used on 3 different parts of that involved gundrill (i have attached picture as well)

teh next thing i worked on was basically we have these mo0dels we get from beong in CATIA and we use inventor t make the process drawing we have to ensure that the drawing smatch up so i did verification of holes, dimensioning, etc and i got to use CATIA, worked will differently assemblies, make this good

worked on sourcing mcmaster parts

was able to talk with one of our vendors about non comfarnce they causes in a part that we sent tem for final coating discussed cause sof the issue and plan going forward, it was for biolase part that is metal tubular part we manufacturer

i also made inventor models and drawings to provide pricing estimat for rafael aerospace to give them pricing on a part they wanted manufacture

I also remade fitting fwd stop for mlg door we only had teh 3d printed model and i remade it my taking measurements in inventor and madde drawings all pictures are attached

the main was the landing gear tubular components

and idk if you can find a way to put tangible numbers in this that show real impact

you can also find out more about the parts, that i worked on, by searching 149t6971-13, 149t6971-11, 149t6971-19 and boeing has some stuff online about it

LeFiell Manufacturing Company — Manufacturing Engineering Intern

V1

- Developed Autodesk Inventor 3D models and process drawings for five Boeing main landing gear door tubular components, translating customer engineering drawings into machining operation sequencing, fixturing and workholding requirements, intermediate part geometry, datum schemes, GD&T, and in-process dimensional tolerances; all five released to production
- Owned the detailed machining progression for precision tubular components on PUMA CNC turning centers, using part geometry, chucking orientation, and tailstock support to define multi setup production of threaded end features, contoured OD geometry, milled flats, and functional hole patterns
- Designed 12 custom thread plugs for RH/LH UNJF-3B internal threads per AS8879, performing assembly-level fit checks and clearance verification in Autodesk Inventor to validate thread engagement and prevent interference during subsequent drilling; iterated tooling geometry with CNC programmers and released fabrication drawings to production.
- Created a gundrill inspection gauge to verify internal radius tangent-point location across three tubular components, integrating three gauging ridges matched to varying bore lengths and replacing three dedicated gauges with one tool
- Reverse-engineered a main landing gear door forward-stop fitting from a physical component and legacy hand drawing, reconstructing critical radii, hole locations, and mating geometry in Autodesk Inventor to create the manufacturing drawing used for production
- Validated Boeing P-8 sonobuoy launch components by comparing CATIA V5 customer models with Autodesk Inventor models and drawings, verifying hole locations, critical dimensional relationships, and feature geometry before production.

V2

LeFiell

- Developed Autodesk Inventor 3D models and process drawings for five Boeing main landing gear tubular components, translating customer drawings into machining operation sequences, intermediate part geometry, datum schemes, defining fixturing and work holding requirements and applying GD&T per ASME Y14.5; all five parts released to production.
- Designed 10 custom thread plugs for RH/LH UNJF-3B threads per AS8879; created Autodesk Inventor assemblies for fit checks to verify thread engagement and drilling clearance, and iterated tooling geometry with CNC programmers.
- Created a gundrill depth gauge for dimensional verification of internal radius tangent point location on three tubular components, using axially spaced gauging ridges matched to varying bore depths and reducing inspection tooling from three gauges to one.
- Reverse engineered a main landing gear door forward stop fitting from a physical component and legacy Boeing hand drawing, using caliper measurements to reconstruct component geometry and build the model in Autodesk Inventor for manufacturing.
- Reduced setup time by ~15% by defining a multi-setup CNC process plan that consolidated first end facing, boring, tapping, and OD contour turning into a single chucking, eliminating an intermediate re-fixturing step before second end machining.
- Validated sonobuoy launch components by cross checking CATIA V5 customer models against Autodesk Inventor models and engineering drawings, verifying dimensional accuracy, hole locations, and feature geometry before production.

- Validated Boeing P-8 sonobuoy launch components by comparing CATIA V5 customer models with Autodesk Inventor 3D models and engineering drawings, verifying hole locations, critical dimensional relationships, and feature geometry before production.

General Dynamics

- Support engineering and procurement execution for COLUMBIA-class SSBNs, working directly with suppliers to resolve fabrication and manufacturability challenges alongside conformance to procurement specifications and PO requirements.
- Review and disposition Vendor Information Requests by validating supplier technical data against engineering drawings, governing welding specifications, and 3D design models in Teamcenter, coordinating technical resolutions and requirement changes across engineering, procurement, and suppliers.

Liaised with company executives and project managers to move projects forward.

Assessed impact of modifications for ongoing projects, providing documentation on required changes and expectations.

Monitored production and equipment for compliance with drawings and specifications.

Analyzed blueprints and made recommendations to improve design, construction means and methods based on field or changing conditions.

Assessed qualifications and experience of contractors and prepared RFP packages in addition to evaluating contractor bid proposals.

Monitored metrics during project execution to maintain compliance with planned costs, schedules and quality goals.

I worked as a Project and Mechanical engineer, expediting manufacturing processes and consulting with builders to ensure contractual compliance.

- Developed Autodesk Inventor process drawings for five Boeing main landing gear door tubular components, interpreting customer engineering drawings to define machining sequence, intermediate geometry, datums, GD&T, and in-process dimensions; all five components were approved and released to production
- Owned the detailed machining progression for precision tubular components on PUMA CNC turning centers, using part geometry, chucking orientation, and tailstock support to define multi setup production of threaded end features, contoured

- Collaborate with a group of 75+ to design and build a hyper-efficient autonomous EV for the Shell Eco-marathon, coming in 5th on the track and winning two off-track awards and \$4500 from the Shell Eco-marathon Americas competition
 - Designed and manufactured a lightweight acceleration pedal and sensor mounts in Autodesk Inventor; analyzed static forces in MATLAB to achieve 25 lbs of pedal feedback force; integrated a linear potentiometer to record position data for testing
 - Led steering system layout for 2026 vehicle and designed a threaded aluminum mounting plate that defines the interface for the rack and pinion, pedal box, and autonomy hardware, engineered for maximum stability and minimal weight
 - Oversee vacuum-infusion and wet layup composite manufacturing, producing high-quality carbon fiber structures that serve as the vehicle's body, alongside machining 50+ custom components using CNC, manual mill, and lathe to support vehicle assembly
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- Built a promotion strategy by aggregating three years of regional sales and delivery-marketplace data in SQL and publishing Tableau dashboards to improve promotional targeting and flag product performance at franchises
 - Created and presented pitch decks to managers outlining go-to-market plans for the promotional campaign, including pilot launches on DoorDash, Uber Eats, and Grubhub; ensured marketing translated into in-store execution across locations, and documented rollout procedures for repeatable execution
 - Advised franchisee owners to set market-appropriate pricing that reflected platform fees and local demand, thereby increasing order volume and profitability, and reviewed performance against forecast each week to drive data-based adjustments
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- Contributing to the MathWorks-sponsored IDS³C Scaled Smart-City Testbed Project, a 1:25-scale research platform to test and validate autonomous vehicle control strategies in urban traffic systems
 - Implemented Arduino/Raspberry Pi hardware across 45 RC vehicles to enable scalable fleet testing and shorten debug cycles by 40%; created an engineering manual covering hardware/software integration & chassis updates, cutting onboarding time by 50%.
 - Built and tested autonomous driving features by integrating scaled road network models in MATLAB, wiring and calibrating force-feedback steering hardware, and driver controls for real-time testing
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- Deliver engaging and informative 90-minute campus tours to prospective students and visitors, highlighting Cornell's academic programs, history, and student life, enhancing Cornell's reputation and outreach efforts

Skills: R, C++, HTML, Java, Python, SQL, Excel, PowerPoint, Office, Policy Writing, Data Analysis & Visualization, Conversational Spanish

Modeling: Autodesk Inventor, Fusion 360, SolidWorks, ANSYS, CFD, FEA, MATLAB, Simulink

- Collaborate with a group of 75+ to design and build a hyper-efficient autonomous EV for the Shell Eco-marathon
- Lead a 5-person chassis subteam through design and structural FEA to deliver a carbon-fiber monocoque, sourced high-density foam cores with Cornell's Digital Design + Fabrication Studio, and managed cross-team systems integration
- Developed the chassis master model in Autodesk Alias and applied ANSYS Fluent CFD to iterate the design into the final aerodynamic geometry, reducing drag by 6% and improving cruise efficiency by ~4% over the previous car design
- Used ANSYS (ACP, Mechanical) and generative design to optimize structural components, cutting weight by 10%
- Designed and manufactured a lightweight accelerator pedal assembly and precision sensor mounts in Autodesk Inventor; implemented a linear potentiometer by connecting to a 5V controller input and calibrated endpoints to a 0.5–4.5 V span, reducing pedal feedback force by 12%
- Contributing to the MathWorks-sponsored IDS³C Scaled Smart-City Testbed Project, a 1:25-scale research platform to test and validate autonomous vehicle control strategies in urban traffic systems
- Overseeing and optimizing wiring systems for autonomous vehicles, to support dependable operation of LiDAR, depth cameras, and onboard processors
- Built and tested autonomous driving features by integrating scaled road network models in MATLAB, wiring and calibrating force-feedback steering hardware, and driver controls for real-time testing

Cornell University Visitor Relations

Tour Guide & Senior Information Specialist

- Deliver engaging and informative 90-minute campus tours to prospective students and visitors, highlighting Cornell's academic programs, history, and student life to foster a welcoming environment
- Served as a primary point of contact at the Visitor Relations Center, addressing inquiries and providing tailored information to support visitors in their decision-making process at the Martin Y Tang Welcome Center

- Demonstrated public speaking and interpersonal skills by effectively communicating complex information to groups of varying sizes, enhancing Cornell's reputation and outreach efforts

I learned Hindi on Duolingo so I could finally speak to my grandparents without my parents translating. The first weeks were humbling, with misplaced plurals and shaky pronunciation, yet I kept the streak. Then one night my nani told me how, as a child in Pakistan, she would sing and dance in the monsoon rain. I followed every detail, asked my question in Hindi, and she laughed and answered back with no translator and no pause. In that moment I felt something click. Later I read her WhatsApp recipe for lamb rogan josh and cooked it exactly her way. Those moments were never about badges. They were about agency. Duolingo did not just teach me words. It gave me access to my own family.

Because of that, I am interested in Duolingo. I am drawn to work where careful measurement and simple design create real progress for real people. As a mechanical engineer, I build systems, test them in the world, and turn data into choices that teams can use. From there, I focus on clarity. Clean data leads to trustworthy analyses and clear success metrics lead to decisions that hold up under scrutiny.

At Duolingo, I want to put that approach to work. For example, I enjoy writing SQL and Python to analyze behavioral data, designing A/B tests with guardrail metrics, and modeling retention and time to unit completion. I also like turning results into short briefs and dashboards so product, learning, and design move together. In the end, I want to help create more of those I understood everything moments for learners everywhere, just like the one that let me hear my grandmother's story the way she tells it.

At Duolingo, I want to put that approach to work. At Flame Broiler I aggregated three years of regional sales and delivery marketplace data in SQL, published Tableau dashboards that flagged underperforming locations, advised market level pricing that reflected platform fees and local demand.

- Contributing to the MathWorks-sponsored IDS³C Scaled Smart-City Testbed Project, a 1:25-scale research platform to test and validate autonomous vehicle control strategies in urban traffic systems
- Optimized wiring systems and tested autonomous features using MATLAB road-network models; wired and calibrated force-feedback steering and driver controls for real-time tests supporting LiDAR, depth cameras, and onboard processors
- Contributing to the MathWorks-sponsored IDS³C Scaled Smart-City Testbed Project, a 1:25-scale research platform to test and validate autonomous vehicle control strategies in urban traffic systems
- Built and tested autonomous driving features by integrating scaled road network models in MATLAB, wiring and calibrating force-feedback steering hardware, and driver controls for real-time testing
- Optimize wiring systems for autonomous vehicles, to support operation of LiDAR, depth cameras, and processors